

Subject Description Form

Subject Code	ITC589
Subject Title	Advanced Textile Processing Technology
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	The subject aims to provide students with an in-depth knowledge on the textile processing technologies. In particular it illustrates the state-of-the-art coloration and finishing technology for imparting fabric properties of aesthetics, function and comfort.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: 1. Demonstrate an extended knowledge and understanding of the principles and technology of textile wet processing. 2. Comprehend the advancement in coloration and finishing technologies and operations. 3. Critically evaluate and solve problems related to coloration and finishing issues. 4. Communicate effectively with professionals in the industries to design and implement systems of textile finishing.
Subject Synopsis/ Indicative Syllabus	Preparation 1. Various stages in textile wet processing 2. Objectives and principles of preparatory processes of textile materials 3. Selection of sequence of processes, batchwise versus continuous mode 4. Water quality requirements, utility and energy sources

	Coloration Dyeing: 1. Classification of dyes according to application 2. Basic principles and mechanisms of dyeing 3. Dyeing textiles at various stages 4. Colour fastness in relation to end uses 5. Machine requirements and selections based on quality and quantity of physical forms of textile materials Printing: 1. Printing methods and style 2. Preparation and machine selection 3. Formulation of printing past including dyestuff, pigment and chemical requirement Finishing method 1. Mechanical finishing 2. Chemical finishing Finishing technology and application 1. Preparatory and physical quality control (heat-setting/tentering etc. for dimensional control, shrinkage resistance, unit weight control, width control, etc.). 2. Finishing technology for appearance and aesthetics (singeing, mercerizing, enzyme treatment, anti-pilling, etc.). 3. Functional finishing technology (abrasion resistant, anti-static, flame retardant, antimicrobial, minimum care (wrinkle free), ultraviolet protection, repellent finishing for water, oil and soil, etc.). 4. Improvement of comfort properties (napping, softness, moisture management, etc.). 5. Nanotechnology application and novel finishing technology
Teaching/Learning Methodology	The subject consists of lectures, seminars and tutorials. Students are required to do investigations on new textile wet processing processes as well as potential developments, then give presentations and they are encouraged to participate in seminar and tutorial discussions.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			
			1	2	3	4
	1. Coursework	50%	✓	✓	✓	✓
	2. Examination	50%	✓	✓	✓	
	Total	100 %				
	In the continuous assessment, we focus on two major attributes: communication skills and problem solving on the technical knowledge. In the seminar and tutorial discussion, interaction among group members is a very good way for enhancing their communication skills in textile wet finishing. Written communication skills are attained through report writing and presentation in the class. In view of technical knowledge, students are assessed through assignments. Meanwhile, problem reasoning and analysis can be developed from group project through searching, selecting, organizing topics and report details. In the examination, we assess students’ learning performance in terms of defining concepts, explain principles and describe and analyze application examples as well as providing rationales for different textile wet processing processes.					
Student Study Effort Expected	Class contact:					
	▪ Lectures			27 Hrs.		
	▪ Seminars/Tutorials			12 Hrs.		
	Other student study effort:					
	▪ Preparation for presentation			30 Hrs.		
	▪ Assignments			30 Hrs.		
	▪ Self-study			36 Hrs.		
	Total student study effort			135 Hrs.		
Reading List and References	1. M. Clark, Handbook of Textile and Industrial Dyeing, Volume 1: Principles, Processes and Types of Dyes, Woodhead Publishing, 2011.					
	2. M. Clark, Handbook of Textile and Industrial Dyeing, Volume 2: Application of Dyes, Woodhead Publishing, 2011.					
	3. J.N. Chakraborty, Fundamentals and Practices in Colouration of Textiles, Woodhead Publishing, 2009.					

	4. B. Christie, Environmental Aspects of Textile Dyeing, Woodhead Publishing, 2007. 5. H. Ujie, Digital Printing of Textiles, Woodhead Publishing, 2006. 6. C. Hawkyard, Synthetic Fibre Dyeing, The Society of Dyers and Colourists, 2004. 7. L.W.C. Miles, Textile Printing, The Society of Dyers and Colourists, 2003. 8. J. Shore, Blends Dyeing, The Society of Dyers and Colourists, 1998. 9. J. Shore, Cellulosics Dyeing, The Society of Dyers and Colourists, 1995. 10. W. Ingamells, Colour for Textiles, The Society of Dyers and Colourists, 1993. 11. A Majumdar, A Das, R Alagirusamy and V K Kothari, Process Control in Textile Manufacturing, Woodhead Publishing Limited, 2013. 12. W.D. Schindler and P J. Hauser, Chemical Finishing of Textiles, Woodhead Publishing Limited, 2004. 13. R12r24 L. Gulraiani, Advances in the Dyeing and Finishing of Technical Textiles, Woodhead Publishing Limited, 2013 14. Derek Heywood, Textile Finishing, Society of Dyers and Colourists, 2003
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